

JOACHIM KULEAFENU

FULL-STACK AI ENGINEER | LLM APPLICATIONS | AGENTIC SYSTEMS

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PROFESSIONAL SUMMARY

Full-Stack AI Engineer with 4+ years of experience building production-grade LLM applications, agentic AI systems, enterprise workflow automation, and cloud-native software platforms. Experienced in multi-agent Retrieval-Augmented Generation, AI-powered SaaS products, RESTful APIs, microservices, and responsive web applications using Golang, Python, FastAPI, React.js, OpenAI APIs, LangChain, AWS, and GCP. Proven ability to own solutions from architecture and development through deployment, testing, observability, and optimization. Recognized with the Great Growth Award and Top Team of the Year for technical leadership, execution, and measurable business impact.

TECHNICAL SKILLS

AI & LLM Engineering: OpenAI APIs, Anthropic Claude, LangChain, LangGraph, LlamaIndex, Retrieval-Augmented Generation, Agentic AI, Multi-Agent Systems, Model Context Protocol, Tool Calling, Structured Outputs, Prompt Engineering, Semantic Search, Vector Databases, Source Attribution, Human-in-the-Loop Workflows, LLM Evaluation, AI Guardrails

Programming & Frameworks: Golang, Python, JavaScript, React.js, FastAPI, PHP, Spring Boot, HTML5, CSS3, Material UI, Pandas, NumPy, Scikit-learn

Cloud & DevOps: AWS Lambda, API Gateway, S3, SQS, CloudWatch, GCP, Docker, Kubernetes, GitHub Actions, Bitbucket Pipelines, CI/CD, Serverless Architecture, Railway

Databases & Data Engineering: PostgreSQL, MySQL, MongoDB, Apache Kafka, Apache Airflow, ETL Pipelines, Data Modeling, Data Ingestion, Data Validation

Architecture & Security: RESTful APIs, Microservices, Event-Driven Architecture, Distributed Systems, Asynchronous Processing, OAuth 2.0, JWT, API Integrations, Multi-Tenant Architecture, System Design

Testing & Delivery: TDD, Unit Testing, Integration Testing, Regression Testing, QA Automation, Application Monitoring, Logging, Error Handling, Code Review, Agile, Scrum, Git, Bitbucket, FFmpeg, Shell Scripting

PROFESSIONAL EXPERIENCE

HUMANEX VENTURES | *Remote*

June 2024 – Present

Software & AI Integration Engineer

- Architect and deploy TalentCast, a cloud-native AI podcast generation platform using Golang, Python, OpenAI APIs, and AWS, automating profile-to-audio content production and reducing manual production effort by approximately 70%.
- Lead the architecture of a multi-agent Retrieval-Augmented Generation platform using LangGraph, LangChain, LlamaIndex, vector databases, and OpenAI APIs to deliver accurate, source-cited answers across enterprise documentation.
- Build retrieval, reasoning, tool-calling, structured-output, and source-attribution workflows that enable employees to access verified organizational knowledge through internal web applications.
- Develop the InsighteX Action Planner, an agentic AI application that analyzes enterprise culture survey data and generates personalized leadership recommendations, reducing analysis time by approximately 60%.
- Design and ship an AI-assisted interview coding platform using Go, AWS Lambda, SQS FIFO, PostgreSQL, React.js, and Material UI, automating free-form response classification across 30+ proprietary assessments.
- Replace fully manual analyst scoring with a human-in-the-loop review and override workflow featuring AI rationales, generated transcripts, question-level synchronized audio playback, entitlement controls, and auditable decision records.
- Develop customer-facing web applications with Golang, React.js, AWS Lambda, and API Gateway, contributing to a 25% improvement in platform performance and user satisfaction.
- Architect a centralized PostgreSQL-backed platform for managing digital licenses and client services, improving operational efficiency by 17% while supporting secure and scalable user workflows.

MINEX360 SERVICES | *Remote*

August 2022 – September 2023

Software Engineer

- Refactor legacy test suites and expand automated unit, integration, and regression testing, reducing execution time by 30% and accelerating delivery across Agile sprints; implement asynchronous processing, validation, logging, authentication, and production monitoring to strengthen reliability.
- Designed and developed scalable RESTful APIs for SmartsApp, supporting secure digital interactions for 500,000+ users across web and mobile platforms.
- Built backend services and data-access layers using Golang, Python, PostgreSQL, and microservice design principles, improving maintainability and horizontal scalability.
- Implemented authentication and authorization workflows using JWT, OAuth 2.0, role-based access controls, token validation, and encrypted data transmission to reduce vulnerability exposure.
- Refactored application code and optimized database queries, caching, and API response handling, reducing server load by approximately 25% and improving system responsiveness.
- Decomposed tightly coupled legacy services into independently deployable microservices, reducing deployment cycle time by approximately 35% and minimizing release dependencies.
- Developed event-driven integrations and asynchronous processing workflows while expanding unit and integration testing, API validation, logging, and exception handling to improve release stability.
- Collaborated with product managers, designers, QA engineers, and developers throughout requirements analysis, technical design, implementation, code review, and deployment.

LUCIDSQUAD | United Kingdom, *Remote*

July 2021 – November 2021

Software Engineer, Contract

- Designed and launched DataHub, a lead-generation and competitive-intelligence platform that identified web technologies used by prospective customers and market competitors.
- Developed Apache Airflow ETL pipelines capable of processing 100,000+ technology and company records per day across automated ingestion, transformation, and validation workflows.
- Optimized pipeline scheduling, data transformation, and error-recovery processes, reducing end-to-end data-processing time by approximately 35%.
- Created a flexible data model supporting company segmentation, technology categorization, competitor analysis, and targeted sales-campaign development.
- Partnered with marketing and product stakeholders to translate business requirements into data-driven features, contributing to an estimated 20% improvement in qualified lead targeting.
- Built backend services and third-party API integrations and mentored junior developers on clean code, Git workflows, testing, debugging, and Agile delivery practices.

SELECTED PROJECT

KASAHQ YOUTUBE RESEARCHER / *Open Source*

yousearcher.kasahq.com

Full-Stack AI Content Production Platform

- Designed and launched an AI-powered content-production platform that transforms an initial video concept into research, scripts, narration, images, captions, and publishing recommendations.
- Orchestrated Anthropic Claude, ElevenLabs, Replicate Flux, Tavily, and FFmpeg through a multi-stage agent workflow that automatically produces 1080 x 1920 short-form videos with word-aligned burned-in captions.
- Integrated YouTube Data and Analytics APIs through per-user OAuth 2.0 and developed a first-48-hour views signal that generates personalized video-publishing playbooks.
- Architected a multi-tenant application using FastAPI, PostgreSQL, React.js, and Railway with tenant-level data isolation, bring-your-own-key management, and modular feature boundaries.
- Protected API credentials and OAuth tokens using Fernet encryption while implementing secure session management, access controls, input validation, and failure-recovery workflows.

EDUCATION

GRAND VALLEY STATE UNIVERSITY | Allendale, Michigan

August 2023 – May 2025

Master of Science in Applied Computer Science | GPA: 4.0/4.0

UNIVERSITY OF ENERGY AND NATURAL RESOURCES

September 2017 – May 2021

Bachelor of Science in Computer Science

LEADERSHIP & AWARDS

- Great Growth Award | 2026
- Top Team of the Year, HumaneX Ventures | 2025
- Graduate Assistant, Grand Valley State University | 2024
- Resident Assistant, Grand Valley State University | 2024